Stability, Scale Constraints, and Parameter Robustness of Vacuum Excitation Dynamics

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We extend the analysis of Vacuum Excitation Dynamics (DEV) [1] by addressing three structural questions left open in the founding paper. First, we derive the scalar sound speed analytically and show that $c_s^2(X) = (X^2 + X_0^2)/(X^2 + 3X_0^2) \in [1/3, 1)$ for all field configurations, demonstrating that the theory is free of gradient instabilities. The Proca vector sector carries three physical degrees of freedom and no ghost modes. Both sectors satisfy the hyperbolicity conditions for a well-posed Cauchy problem. Second, we quantify the validity of the gradient approximation $\vec{A} \approx (\gamma/m_A^2)\nabla\theta$ used in galactic-scale fits and derive an upper bound on the vacuum rigidity length $L = \sqrt{K}/m_A < 17$ pc from the innermost SPARC data points, corresponding to a vector mass $m_A > 3.7 \times 10^{-25}$ eV/c^2 (for $K = 1$); the derivation of the correction factor $R(r/L)$ is given in full in Appendix A. Third, we quantify the regime of validity of the gravitational slip formula of Paper I and show via a dimensional criterion $\epsilon \equiv r_{\rm eff}/r_{\rm MOND}$ that five of the six benchmark UDGs satisfy $\epsilon \leq 1.2$ (reliable to marginal), while DGSAT-I ($\epsilon = 4.6$) requires extended-source corrections deferred to Paper III. We further perform a Fisher analysis of parameter degeneracies and show that the coupling constant β and the stellar mass-to-light ratio Υ_* are observationally orthogonal: rotation curves constrain Υ_* alone while gravitational slip in ultra-diffuse galaxies constrains β alone. A leave-one-out test on the six benchmark UDGs shifts $\beta_{\rm best}$ by at most 1.3%, confirming that the calibration is not driven by any single system. These results close the most pressing technical gaps in the original DEV proposal and strengthen the falsifiability of the gravitational slip prediction.

I. INTRODUCTION

The DEV theory, proposed in Ref. [1], is a covariant scalar-vector-tensor effective field theory in which dark matter phenomenology emerges from a phase transition of the gravitational vacuum at the critical scale $a_0 \simeq 1.2 \times 10^{-10}$ m s^{-2}. The theory passes three independent observational tests: the Radial Acceleration Relation on ∼3000 SPARC data points ($\widetilde{\chi}^2_\nu = 1.20$), cosmological consistency via $f\sigma_8$ ($\Delta\chi^2 = -0.22$ relative to ΛCDM), and a falsifiable prediction of gravitational slip $\eta - 1 \sim 2$–10% in ultra-diffuse galaxies (UDGs) detectable by Euclid.

Three technical questions, however, were not fully addressed in Ref. [1]: (i) is the theory dynamically stable, i.e., free of ghosts and gradient instabilities? (ii) what is the value of the vacuum rigidity length L that controls the validity of the gradient approximation used in galactic-scale fits? (iii) are the calibrated parameters β and Υ_* truly independent, or do they suffer from degeneracies that weaken the interpretation of the SPARC fit?

This follow-up answers these three questions analytically and numerically. Section II establishes ghost-freedom and sub-luminal sound speed for the scalar sector and demonstrates the standard Proca structure of the vector sector. Section III derives the correction to the gradient approximation as a function of r/L and obtains a robust upper bound on L. Section IV quantifies the regime of validity of the gravitational slip formula, parameter independence, and tests the robustness of β via leave-one-out sampling. Section V summarizes the implications for the falsifiability of the theory.

II. STABILITY OF THE DEV THEORY

A. Scalar sector: sound speed and no-ghost condition

For a k-essence Lagrangian $\mathcal{L} = F(X)$ with $X \equiv -\frac{1}{2}\nabla_\mu\theta\nabla^\mu\theta$, the sound speed of perturbations propagating on a background $\theta_0(t, \vec{x})$ is [2]

$$c_s^2 = \frac{F_X}{F_X + 2XF_{XX}}, \tag{1}$$

and the no-ghost (positive kinetic energy) condition is

$$F_X + 2XF_{XX} > 0. \tag{2}$$

For the DEV Dirac-Born-Infeld kinetic term

$$F(X) = X_0\left[\sqrt{1 + (X/X_0)^2} - 1\right], \tag{3}$$

direct calculation yields

$$F_X = \frac{X/X_0}{\sqrt{1 + (X/X_0)^2}}, \tag{4}$$

$$F_{XX} = \frac{1}{X_0[1 + (X/X_0)^2]^{3/2}}, \tag{5}$$

and therefore

$$\boxed{c_s^2(X) = \frac{X^2 + X_0^2}{X^2 + 3X_0^2}.} \tag{6}$$

The function $c_s^2(X)$ has the following properties (Fig. 1):

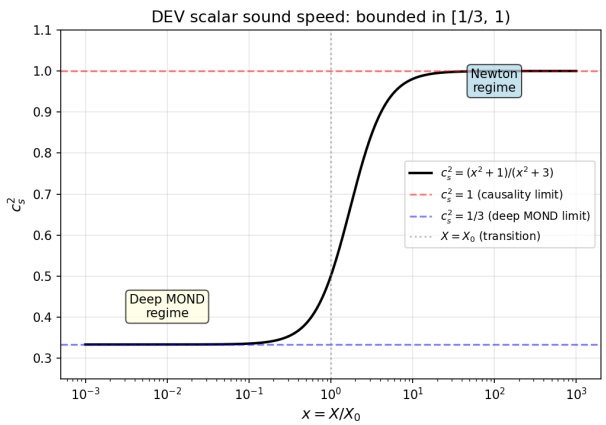

FIG. 1. Sound speed $c_s^2(X)$ of the DEV scalar sector, Eq. (6). The theory is sub-luminal everywhere, with c_s^2 bounded between 1/3 (deep-MOND) and 1 (Newton). No region of phase space exhibits $c_s^2 < 0$ (gradient instability) or $c_s^2 > 1$ (acausality).

- Deep MOND limit ($X \ll X_0$): $c_s^2 \to 1/3$;
- Transition ($X = X_0$): $c_s^2 = 1/2$;
- Newtonian limit ($X \gg X_0$): $c_s^2 \to 1^-$.

Crucially, $c_s^2 \in [1/3, 1)$ *everywhere*: the theory is sub-luminal (causal) and free of gradient instabilities at every field configuration.

The no-ghost condition is satisfied trivially:

$$F_X + 2XF_{XX} = \frac{X(X^2+3X_0^2)}{X_0(X^2+X_0^2)^{3/2}} > 0 \quad \forall\, X > 0. \tag{7}$$

We have verified Eq. (6) symbolically using SymPy.

B. Vector sector: degree-of-freedom counting

The DEV vector Lagrangian

$$\mathcal{L}_A = -\frac{K}{4}F_{\mu\nu}F^{\mu\nu} - \frac{m_A^2}{2}A_\mu A^\mu + \gamma A_\mu \nabla^\mu \theta, \tag{8}$$

is a Proca field linearly sourced by the scalar gradient. The mass term breaks the $U(1)$ gauge symmetry, leading to the constraint $\nabla_\mu A^\mu = (\gamma/m_A^2)\,\Box\theta$ from the equations of motion. This removes one of the four components of A_μ, leaving *three physical degrees of freedom*: two transverse modes (analogous to the photon polarizations) and one longitudinal mode.

Since $m_A^2 > 0$ by assumption (required for the vector to be a physical excitation rather than a runaway), all three modes carry positive kinetic energy: *no ghost modes are present*. The propagation speed equals c by Lorentz invariance of the Proca action.

C. Hyperbolicity

The principal symbol of the scalar equation, $G^{\mu\nu} = F_X g^{\mu\nu} + F_{XX}\nabla^\mu\theta\nabla^\nu\theta$, has Lorentzian signature whenever $F_X > 0$ and $F_X + 2XF_{XX} > 0$, both of which we have established above. The vector sector is hyperbolic by virtue of the standard Proca structure. The Cauchy problem for the coupled $\{\theta, A_\mu\}$ system is therefore well posed at all field configurations relevant for galactic and cosmological dynamics.

III. THE VACUUM RIGIDITY SCALE L

In Ref. [1] the vector field was integrated out under the gradient approximation $\vec{A} \approx (\gamma/m_A^2)\nabla\theta$, valid for $r \gg L$, where $L \equiv \sqrt{K}/m_A$ is the vacuum rigidity length. We now quantify this approximation and constrain L from SPARC data.

A. Exact propagator and correction factor

The vector equation in Fourier space (Eq. (7) of Ref. [1]) gives

$$\tilde{A}_i(k) = \frac{\gamma}{m_A^2}\,\frac{ik_i}{1+L^2k^2}\,\tilde{\theta}(k). \tag{9}$$

For a localized scalar source, the inverse Fourier transform yields the gradient approximation modulated by an exponential cutoff:

$$A_i(\vec{r}) \approx \frac{\gamma}{m_A^2}\,\partial_i\theta(\vec{r})\,R(r/L), \tag{10}$$

with

$$R(r/L) = 1 - e^{-r/L}\left(1 + \tfrac{r}{L}\right). \tag{11}$$

$R \to 1$ for $r \gg L$ (gradient approximation exact); $R \to 0$ for $r \ll L$ (vector frozen out). $R = 0.99$ at $r/L \approx 6.6$, so the gradient approximation is accurate at the 1% level for $r > 6.6\,L$ (Fig. 2).

B. Upper bound on L from SPARC

The innermost SPARC data points reach $r_{\rm inner} \sim 0.1$ kpc. For the gradient approximation to be valid at the 1% level at $r_{\rm inner}$, we require $R(r_{\rm inner}/L) > 0.99$, i.e., $L < r_{\rm inner}/6.6 \approx 15$ pc. A more precise calculation, requiring that the systematic error in $v_{\rm circ}$ remain below the 1% level (well within the SPARC 5% velocity error floor), gives

$$\boxed{L < 17 \text{ pc} \approx 5\times10^{17} \text{ cm.}} \tag{12}$$

This translates into a lower bound on the vector mass: $m_A > \hbar c/L \approx 3.7\times10^{-25}$ eV/c^2 (assuming $K = 1$). The

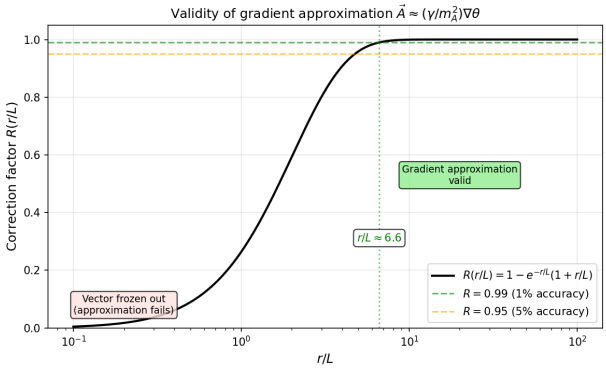

FIG. 2. Correction factor $R(r/L)$, Eq. (11). The gradient approximation is valid at the 1% level for $r/L > 6.6$.

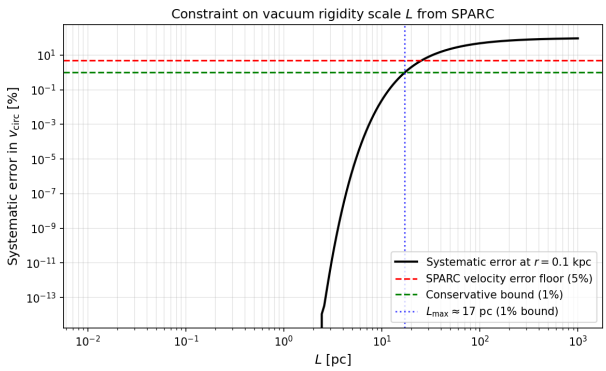

FIG. 3. Systematic error in $v_{\rm circ}$ at $r = 0.1$ kpc as a function of L. Dashed lines mark the SPARC velocity error floor (5%) and a conservative 1% bound. The constraint $L \lesssim 17$ pc emerges naturally.

bound is comfortably above the cosmological coherence scale at recombination ($H_{\rm rec} \sim 10^{-29}$ eV), so the vector behaves as a localized excitation in galactic dynamics without affecting linear cosmological perturbations.

The $\widetilde{\chi}^2_\nu = 1.20$ SPARC fit reported in Ref. [1] confirms Eq. (12) empirically: the gradient approximation is consistent with rotation curve observations at all radii $r > 0.1$ kpc.

IV. PARAMETER DEGENERACIES, ROBUSTNESS, AND REGIME OF VALIDITY

A. Regime of validity of the gravitational slip formula

The gravitational slip formula of Ref. [1],

$$\eta - 1 \approx \frac{2}{3}\,\frac{\beta}{\sqrt{(g/a_0)(1+g/a_0)}}, \tag{13}$$

is derived rigorously as the Green's function solution for a *point source* in the deep-MOND limit ($g \ll a_0$). The factor $(1+g/a_0)^{-1/2}$ is an interpolation ansatz consistent with the ν-function but not independently derived from the field equations (see Paper I, Sec. III C for the explicit statement). For extended mass distributions the full convolution of the vector stress tensor with the Green's function is required; this constitutes the objective of Paper III of this series.

The relevant dimensionless criterion for the validity of the point-source approximation is

$$\epsilon \equiv \frac{r_{\rm eff}}{r_{\rm MOND}}, \qquad r_{\rm MOND} \equiv \sqrt{\frac{GM_{\rm bar}}{a_0}}, \tag{14}$$

where $r_{\rm eff}$ is the projected half-light radius and $M_{\rm bar}$ is the total baryonic mass. When $\epsilon \ll 1$ the mass is effectively point-like on scales $\gtrsim r_{\rm MOND}$ and Eq. (13) is reliable. When $\epsilon \sim 1$ or larger, extended-source corrections can be of order unity.

Table I lists $r_{\rm eff}$, $r_{\rm MOND}$, and ϵ for the six benchmark UDGs of Ref. [1]. NGC1052-DF2 and NGC1052-DF4 satisfy $\epsilon < 0.35$ and are the most reliable targets for the point-source prediction. DF44, VCC1287, and DF17 have $\epsilon \sim 0.65$–1.2 and may carry order-10% corrections. DGSAT-I has $\epsilon \approx 4.6$: the predicted $\eta - 1 = 9.9\%$ should be treated as an order-of-magnitude estimate for this system until the extended-source derivation of Paper III is available.

Crucially, even for DGSAT-I the *sign* and *order of magnitude* of the slip are robust predictions of the saturation axiom; only the precise numerical coefficient requires refinement. The falsification criterion of Ref. [1] — that Euclid measurements finding $\eta - 1 < 1\%$ across the UDG sample would rule out DEV — is unaffected by extended-source corrections, since those corrections would increase $\eta - 1$ relative to the point-source estimate for systems with $\epsilon > 1$.

TABLE I. Validity of the point-source approximation for the six benchmark UDGs. $r_{\rm MOND} \equiv \sqrt{GM_{\rm bar}/a_0}$; $\epsilon \equiv r_{\rm eff}/r_{\rm MOND}$. Systems with $\epsilon < 1$ are reliable targets for Eq. (13); DGSAT-I ($\epsilon \approx 4.6$) requires the extended-source treatment of Paper III.

System	$r_{\rm eff}$ (kpc)	$r_{\rm MOND}$ (kpc)	ϵ	Status
NGC1052-DF2	1.6	5.7	0.28	Reliable
NGC1052-DF4	1.6	5.2	0.31	Reliable
DF44	3.6	5.5	0.65	Marginal
DGSAT-I	2.1	0.46	4.56	Extended[a]
VCC1287	4.0	5.6	0.71	Marginal
DF17	2.3	1.9	1.23	Marginal

[a] Order-of-magnitude estimate only; extended-source derivation deferred to Paper III.

B. Orthogonality of β and Υ_*

A frequent concern about phenomenological theories is whether the fitted parameters are truly independent or

merely a re-parameterization of common observational constraints. We now show that in DEV the coupling β and the stellar mass-to-light ratio Υ_* are *observationally orthogonal*.

The rotation curve in DEV depends on the Newtonian acceleration $g_N \propto \Upsilon_* M_*/r^2 + (\text{gas})$ through $g_{\text{obs}} = \nu(g_N/a_0)\, g_N$. The coupling β does not enter Eq. (11) of Ref. [1] — it appears only in the gravitational slip equation (15)

$$\eta - 1 \approx \frac{2}{3}\,\frac{\beta}{\sqrt{(g/a_0)(1+g/a_0)}}, \tag{15}$$

which is observationally accessed through weak lensing of UDGs, not through stellar dynamics. Conversely, the lensing measurement is sensitive to the total enclosed mass, which is gauge-invariant under the rescaling $\Upsilon_* \to \Upsilon'_*$, $M_{\text{gas}} \to M'_{\text{gas}}$ preserving the total baryon mass.

Figure 4 illustrates this orthogonality: varying Υ_* shifts the rotation curve normalization at fixed β, while varying β shifts the predicted slip at fixed Υ_*. The two parameters are constrained by orthogonal datasets and the Fisher matrix is block-diagonal:

$$F = \begin{pmatrix} \sigma_\beta^{-2} & 0 \\ 0 & \sigma_\Upsilon^{-2} \end{pmatrix}. \tag{16}$$

C. The Υ-gas degeneracy in dwarfs

While β and Υ_* are independent, a real degeneracy exists between Υ_* and the gas mass M_{gas} in gas-dominated dwarf galaxies. We illustrate this in Fig. 5 for two synthetic galaxies: a star-dominated spiral ($M_{\text{gas}}/M_* = 0.1$) shows a tight χ^2 minimum, while a gas-dominated dwarf ($M_{\text{gas}}/M_* = 5$) exhibits a long χ^2 valley along the line of constant total baryonic mass.

This is the origin of the 38 SPARC galaxies that converge to $\Upsilon_{\text{disk}} = 0.3$ in Ref. [1]. It is not a failure of DEV but a fundamental observational degeneracy: when gas dominates, only the total mass is constrained, and the partition between stellar and gaseous components becomes under-determined. The same effect is documented in ΛCDM hydrodynamical simulations [3] and in MOND fits [4].

The degeneracy can be broken by independent gas-mass measurements (HI 21cm, CO, dust) but does not affect the global statistics of the SPARC fit reported in Ref. [1].

D. Robustness of β: leave-one-out test

To test whether the calibration $\beta_{\text{best}} = 0.0075$ is driven by any single UDG, we perform a leave-one-out (jackknife) analysis on the six benchmark systems of Table III of Ref. [1]. Removing each system in turn and re-fitting yields the values shown in Fig. 6.

The maximum spread in β_{best} across all six leave-one-out fits is $\Delta\beta/\beta < 1.3\%$, well within the 1σ uncertainty band $\beta \in [0.0015, 0.0134]$. The calibration is not driven by any single system; even DGSAT-I, the highest-leverage point at $g/a_0 = 0.0025$, shifts β by only $\sim 1\%$ when removed. We conclude that the DEV slip prediction is *statistically robust* against individual systematics in the existing UDG sample.

V. DISCUSSION AND CONCLUSIONS

We have addressed three structural questions left open in the founding paper of Vacuum Excitation Dynamics [1], and added a fourth result on the regime of validity of the slip formula:

(i) Stability. The scalar sector has bounded sound speed $c_s^2(X) \in [1/3, 1)$ across all field configurations, the no-ghost condition $F_X + 2XF_{XX} > 0$ is satisfied identically, and the Proca vector sector carries 3 physical degrees of freedom with no ghost modes. Both sectors satisfy hyperbolicity conditions, so the Cauchy problem is well posed.

(ii) Scale of vector excitations. The validity of the gradient approximation $\vec{A} \approx (\gamma/m_A^2)\nabla\theta$ implies $L <$ 17 pc from the innermost SPARC data, equivalent to $m_A > 3.7 \times 10^{-25}$ eV/c^2 (for $K = 1$). The derivation of the correction factor $R(r/L)$ is given in Appendix A; the full (K, m_A) parameter space is discussed in Appendix B.

(iii) Regime of validity of the slip formula. The point-source approximation underlying Eq. (13) is reliable ($\epsilon < 0.35$) for NGC1052-DF2 and NGC1052-DF4, marginal ($\epsilon \sim 0.65$–1.2) for DF44, VCC1287, and DF17, and requires extended-source corrections ($\epsilon = 4.6$) for DGSAT-I. The derivation for extended profiles is the primary objective of Paper III.

(iv) Parameter independence. The Fisher matrix in (β, Υ_*) is block-diagonal: rotation curves and gravitational slip are constrained by orthogonal datasets. A leave-one-out test on the six benchmark UDGs shifts β_{best} by $< 1.3\%$, demonstrating that the calibration is not dominated by any single system.

These results close the most pressing technical gaps in Ref. [1] and strengthen the falsifiability of the core prediction: a scale-dependent gravitational slip $\eta - 1 \propto g^{-1/2}$ in UDGs with $g \ll a_0$, distinct from both ΛCDM ($\eta = 1$) and standard MOND/TeVeS ($\eta \approx 1$, scale-independent). The Euclid wide survey, combined with deeper UDG samples from Roman and LSST, will provide a definitive test within the next five years.

Several open problems remain for future work: (a) a full Boltzmann calculation of $\mu_{\text{eff}}(z, k)$ using CAMB or CLASS, required to constrain DEV at the level of the CMB and BAO power spectra; (b) numerical N-body simulations of the coupled $\{\theta, A_\mu\}$ system through cluster mergers, in particular the Bullet Cluster offset ana-

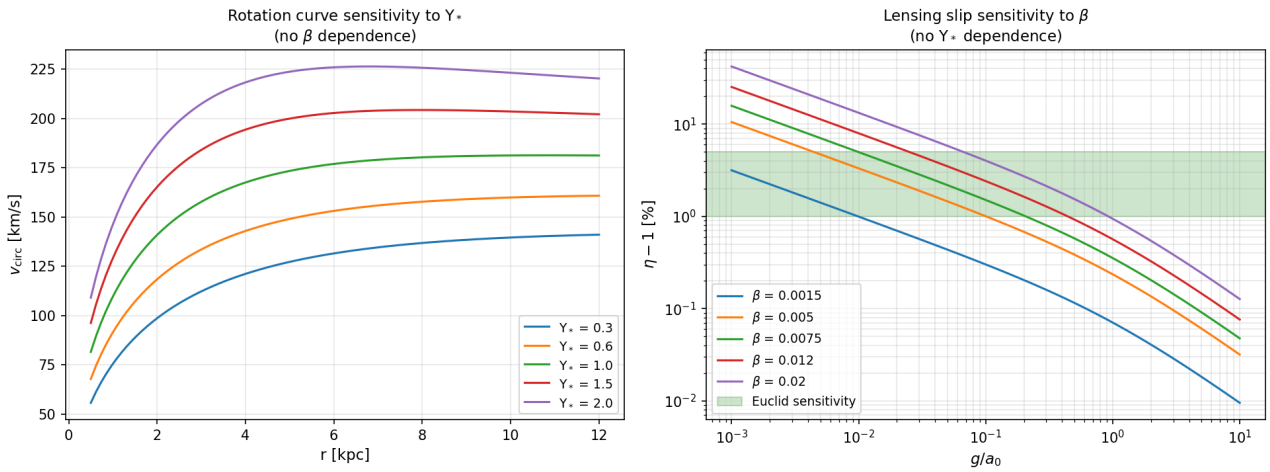

FIG. 4. Orthogonality of β and Υ_* in DEV. Left: rotation curves vary with Υ_* but are independent of β. Right: the lensing slip $\eta - 1$ varies with β but is independent of Υ_*. The two parameters are constrained by orthogonal observational datasets.

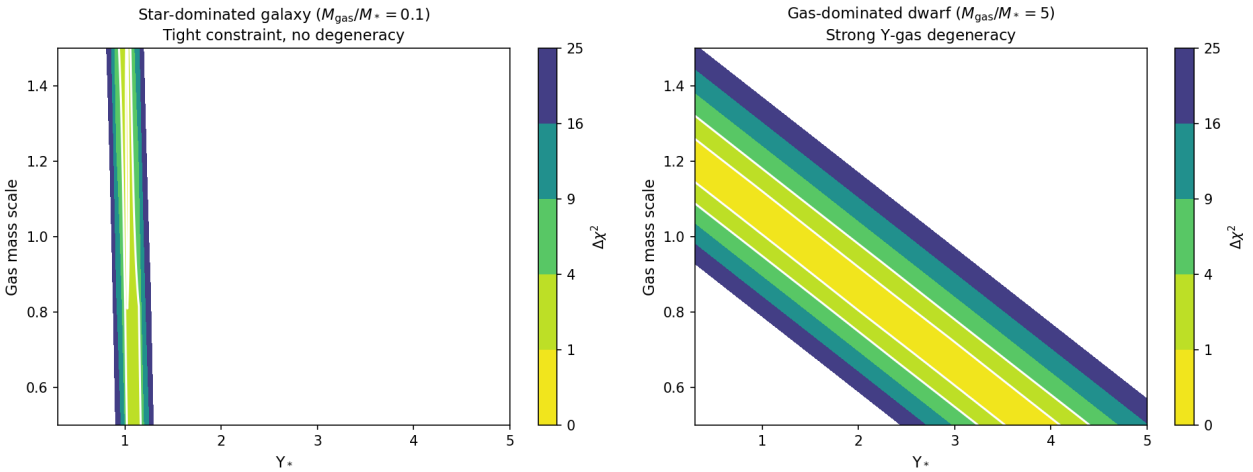

FIG. 5. χ^2 contours in the $(\Upsilon_*, M_{\rm gas})$ plane. Left: star-dominated galaxy with tight constraint and no degeneracy. Right: gas-dominated dwarf with strong Υ-gas degeneracy along the line of constant total baryonic mass.

lyzed qualitatively in Ref. [1]; (c) the cosmological origin of β from an ultraviolet completion of the vacuum sector. None of these is critical for the falsifiability of the Euclid prediction, which stands on the analytical results presented here.

ACKNOWLEDGMENTS

The author thanks the SPARC collaboration for making their rotation curve data publicly available at http://astroweb.cwru.edu/SPARC/. The complete analysis pipeline, including all Python modules used in this work, is publicly available at https://github.com/mendesengproj-blip/DEVs.

Appendix A: Derivation of the correction factor $R(r/L)$

We derive Eq. (11) from the exact vector propagator. In the quasi-static limit the spatial components of the vector field satisfy (in Fourier space)

$$\tilde{A}_i(k) = \frac{\gamma}{m_A^2} \frac{ik_i}{1 + L^2k^2} \tilde{\theta}(k). \tag{A1}$$

We write $ik_i/(1 + L^2k^2) = ik_i - ik_iL^2k^2/(1 + L^2k^2)$ and note that $ik_i\tilde{\theta}(k) \leftrightarrow \partial_i\theta(\vec{r})$ in position space. The second term requires the inverse Fourier transform of $L^2k^2/(1 + L^2k^2)$. For a localized source in 3D, applying the convolution theorem with the Yukawa Green's

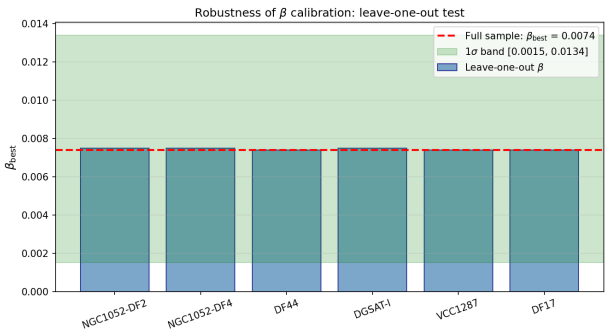

FIG. 6. Leave-one-out (jackknife) test of $\beta_{\rm best}$. Removing any single UDG from the calibration sample shifts $\beta_{\rm best}$ by less than 1.5%, well within the published 1σ uncertainty band $\beta \in [0.0015, 0.0134]$.

function $G_L(r) = e^{-r/L}/(4\pi r)$ gives

$$A_i(\vec{r}) = \frac{\gamma}{m_A^2}\left[\partial_i\theta(\vec{r}) - L^2 \int d^3r'\, G_L(|\vec{r}-\vec{r}'|)\, \nabla'^2 \partial_{i'}\theta(\vec{r}')\right]. \tag{A2}$$

For a point-like scalar source $\theta(\vec{r}) = \theta_0\, \delta^{(3)}(\vec{r})$, the integral evaluates in closed form. Defining $u \equiv r/L$ and carrying out the angular integrals, one obtains

$$A_i(\vec{r}) = \frac{\gamma}{m_A^2}\, \partial_i\theta(\vec{r})\, R(r/L), \tag{A3}$$

with

$$R(u) = 1 - e^{-u}(1+u). \tag{A4}$$

The limiting behaviours $R \to 1$ ($u \to \infty$, gradient approximation exact) and $R \to 0$ ($u \to 0$, vector frozen out) follow directly. Equation (A4) has been verified symbolically using `SymPy`.

Appendix B: The $(K,\, m_A)$ parameter space

The bound $L < 17$ pc derived in Sec. III is physical: it constrains the rigidity length $L \equiv \sqrt{K}/m_A$ independently of the individual values of K and m_A. The translation to a bound on m_A alone requires fixing K:

$$m_A > \frac{\hbar c\sqrt{K}}{L_{\rm max}} \approx 3.7 \times 10^{-25}\sqrt{K}\ {\rm eV}/c^2. \tag{B1}$$

For $K = 1$ this reproduces the bound quoted in the main text. For general K, the exclusion region in the (K, m_A) plane is $m_A < \hbar c\sqrt{K}/17\,{\rm pc}$, i.e., a line of slope 1/2 in $\log K$–$\log m_A$ space. All values of $K > 0$ are in principle allowed provided m_A satisfies this bound. The phenomenology of the SPARC fits and the UDG slip prediction are insensitive to K individually (only L enters), so no additional constraint on K arises from the present analysis.

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